

Software Requirements Specification

Integrated Large-Scale Parking Management System

Version 3.0

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Revision History

Name	Date	Reason for Changes	Version
Moczowski Kacper	07.05.2026	Initial version	1.0
Moczowski Kacper	14.05.2026	Added UCD	2.0
Moczowski Kacper	05.06.2026	Added UML Class Diagram	3.0

1 Introduction

1.1 Purpose

The purpose of the system is to manage parking operations and traffic control in large-scale, multi-purpose environments such as cities, boroughs, company hubs, logistics centers, and shopping centers. The system is intended to improve parking efficiency, reduce traffic congestion, and provide real-time monitoring and management capabilities.

1.2 Scope

The Integrated Large-Scale Parking Management System is designed to provide centralized parking management and traffic monitoring functionalities across multiple parking zones and facilities. The system enables users to locate available parking spaces, reserve spots, perform online payments, and receive navigation assistance.

The platform also supports administrators by providing real-time monitoring, parking analytics, traffic flow management, user management, and reporting tools. The system integrates with external services and devices such as IoT parking sensors, license plate recognition systems, and payment gateways.

The primary objective of the system is to optimize parking utilization, reduce vehicle congestion, and improve the overall user experience in urban and commercial environments.

1.3 Audience

This document is intended for multiple groups involved in the development, deployment, testing, and usage of the system. The primary audience includes software developers, system architects, project managers, testers, stakeholders, and future maintainers of the application.

The document is organized into sections describing the overall system architecture, functional and non-functional requirements, external interfaces, constraints, and system features. Readers may focus on sections relevant to their responsibilities and interests.

2 Overall Description

2.1 Product Perspective

The Integrated Large-Scale Parking Management System is a new, self-contained software solution designed to improve parking management and traffic control processes in urban and commercial environments. The system may serve as a replacement for traditional parking management mechanisms, such as parking meters, paper tickets, and manually operated parking control systems.

The platform integrates modern technologies including IoT sensors, automated license plate recognition, online payment systems, and centralized traffic monitoring services. The system is intended to provide efficient and scalable parking infrastructure management for cities, logistics centers, company campuses, shopping centers, and other large-scale facilities.

2.2 User Classes

The system is intended for multiple categories of users with different responsibilities and access privileges.

- **Drivers** - users searching for available parking spaces, making reservations, performing payments, and managing parking sessions through mobile or web applications.
- **Parking Administrators** - personnel responsible for managing parking zones, monitoring occupancy, configuring parking rules, and generating reports.
- **System Operators** - technical staff responsible for maintaining system availability, monitoring infrastructure, and resolving operational issues.
- **Payment Providers** - external financial services responsible for secure payment processing and transaction verification.
- **IoT Sensor Network** - integrated hardware devices used for detecting vehicle presence, monitoring parking occupancy, and transmitting real-time data to the system.
- **City Traffic Management Systems** - external municipal systems integrated with the platform to improve traffic analysis and congestion management.

The primary user class of the system consists of drivers and parking administrators, as they interact directly with the platform on a daily basis.

2.3 Constraints

The system development and deployment process is subject to several technical and operational constraints.

- High hardware and infrastructure requirements due to real-time data processing and large-scale deployment.
- Significant server load caused by continuous communication with IoT devices and large numbers of concurrent users.
- Maintenance and operational costs related to server infrastructure, sensor networks, and system monitoring.
- Dependence on stable network connectivity for real-time synchronization and communication.
- Compliance with security and privacy regulations with respect to user data and payment processing.
- Compatibility requirements with external systems such as payment gateways, traffic management services, and license plate recognition technologies.
- Scalability requirements to support future expansion of parking zones and increased user traffic.

3 Functional Requirements

- **FR-01:** The system shall allow users to create and manage accounts free of charge.
- **FR-02:** The system shall display available parking spaces in real time.
- **FR-03:** The system shall allow users to reserve parking spaces before arrival.
- **FR-04:** The system shall support online payment processing for parking services.
- **FR-05:** The system shall automatically recognize vehicle license plates using integrated camera systems.
- **FR-06:** The system shall provide navigation assistance directing drivers to selected parking locations.
- **FR-07:** The system shall allow parking administrators to monitor parking occupancy and traffic conditions in real time.
- **FR-08:** The system shall generate usage statistics and analytical reports for administrators.
- **FR-09:** The system shall maintain secure authentication and authorization mechanisms for all registered users.
- **FR-10:** The system shall store parking history and transaction records for registered users.
- **FR-11:** The system shall detect invalid or unauthorized parking sessions and notify administrators automatically.
- **FR-12:** The system shall remain operational in case of temporary network interruptions and synchronize data after reconnection.
- **FR-13:** The system shall provide multilingual user interface support.
- **FR-14:** The system shall provide APIs for integration with external traffic management and payment systems.

4 Non-Functional Requirements

4.1 Performance Requirements

- **NFR-P1:** The system shall process and reflect parking state changes within 15 seconds under normal operating conditions.
- **NFR-P2:** The system shall support a minimum of 10,000 concurrent active users without significant performance degradation.

4.2 Scalability Requirements

- **NFR-S1:** The system architecture shall support horizontal scaling to accommodate additional parking zones, users, and IoT devices without major redesign.
- **NFR-S2:** The system shall allow incremental expansion of parking areas without service downtime.

4.3 Availability and Reliability

- **NFR-A1:** The system shall achieve a minimum uptime of 99.5% excluding planned maintenance.
- **NFR-A2:** The system shall ensure eventual consistency across distributed components within 20 seconds.
- **NFR-A3:** The system shall continue operating in a degraded mode during partial system or network failures.

4.4 Security Requirements

- **NFR-Sec1:** The system shall encrypt all sensitive data in transit and at rest using industry-standard encryption protocols.
- **NFR-Sec2:** The system shall implement authentication and role-based access control for all user classes.

4.5 Privacy Requirements

- **NFR-PR1:** The system shall comply with applicable data protection regulations (e.g., GDPR) regarding user data storage and processing.

4.6 Usability Requirements

- **NFR-U1:** The system shall provide an intuitive interface allowing users to complete key actions (search, reserve, pay) within 3 steps.
- **NFR-U2:** The system shall provide a consistent user interface across all supported platforms.

4.7 Maintainability Requirements

- **NFR-M1:** The system shall be modular to allow independent development, testing, and deployment of components.
- **NFR-M2:** The system shall provide logging and monitoring mechanisms for debugging and performance analysis.

4.8 Interoperability Requirements

- **NFR-I1:** The system shall support integration with external services such as payment gateways, mapping services, and traffic management systems via standardized APIs.

4.9 Portability Requirements

- **NFR-PO1:** The system shall be compatible with at least three major operating systems, including Android, iOS, and Windows-based environments.

4.10 Data Integrity Requirements

- **NFR-D1:** The system shall ensure that no more than 0.1% of transactions result in inconsistent or lost data under normal operation.

5 External Interface Requirements

5.1 User Interfaces Overview

The Integrated Large-Scale Parking Management System shall provide a modern graphical user interface (GUI) accessible via web and mobile applications. The system shall allow users to interact with parking services through intuitive visual components such as buttons, maps, lists, and forms.

From the user's perspective, the system shall support the following high-level interactions:

- Searching for available parking spaces using a map-based or list-based view.
- Viewing real-time parking availability and occupancy status.
- Reserving a parking spot before arrival.
- Starting and ending a parking session automatically or manually.
- Performing online payments for parking services.
- Receiving notifications regarding parking status, time expiration, or system alerts.

The system shall provide immediate feedback to the user after each action, such as confirmation messages, error messages, or updated parking status. For example:

- Confirmation of successful reservation.
- Warning when no parking spaces are available.
- Notification when parking time is about to expire.

The interface shall be responsive and adapted to different screen sizes (desktop, tablet, mobile). The UI shall follow a consistent design system, including:

- Standard navigation bar available on all screens.
- Consistent placement of key actions (search, reserve, pay).
- Unified error and success message formatting.
- Clear visual indicators for parking availability (e.g., color-coded markers).

User interaction shall primarily be based on clicking/tapping UI elements, selecting map locations, and entering minimal text input (e.g., account login or payment details).

5.2 Hardware Interfaces

The system shall interface with various hardware components used in parking infrastructure. These include:

- IoT parking sensors for detecting vehicle presence and occupancy status.
- Cameras equipped with Automatic Number Plate Recognition (ANPR) for vehicle identification.
- Parking barriers and gate control systems for entry and exit management.
- Payment terminals (optional) for on-site transactions.

Hardware devices shall transmit real-time data to the system regarding vehicle entry, exit, and parking space occupancy. Communication between hardware and software components shall be performed using standard network protocols such as HTTP, MQTT, or WebSocket, depending on the device capabilities.

5.3 Software Interfaces

The system shall integrate with external software components and services to ensure full functionality. These include:

- Payment gateway services (e.g., online payment providers) for processing transactions securely.
- Mapping and geolocation services for displaying parking locations and navigation routes.
- Traffic management systems for integration with city-level infrastructure.
- Authentication services for user login and identity management (optional third-party providers).
- Database management systems for persistent storage of user, parking, and transaction data.

The system shall exchange data with external components using secure RESTful APIs and standardized data formats such as JSON. Incoming data may include:

- Parking sensor updates (occupancy status).
- Payment confirmations.
- Vehicle identification data.

Outgoing data may include:

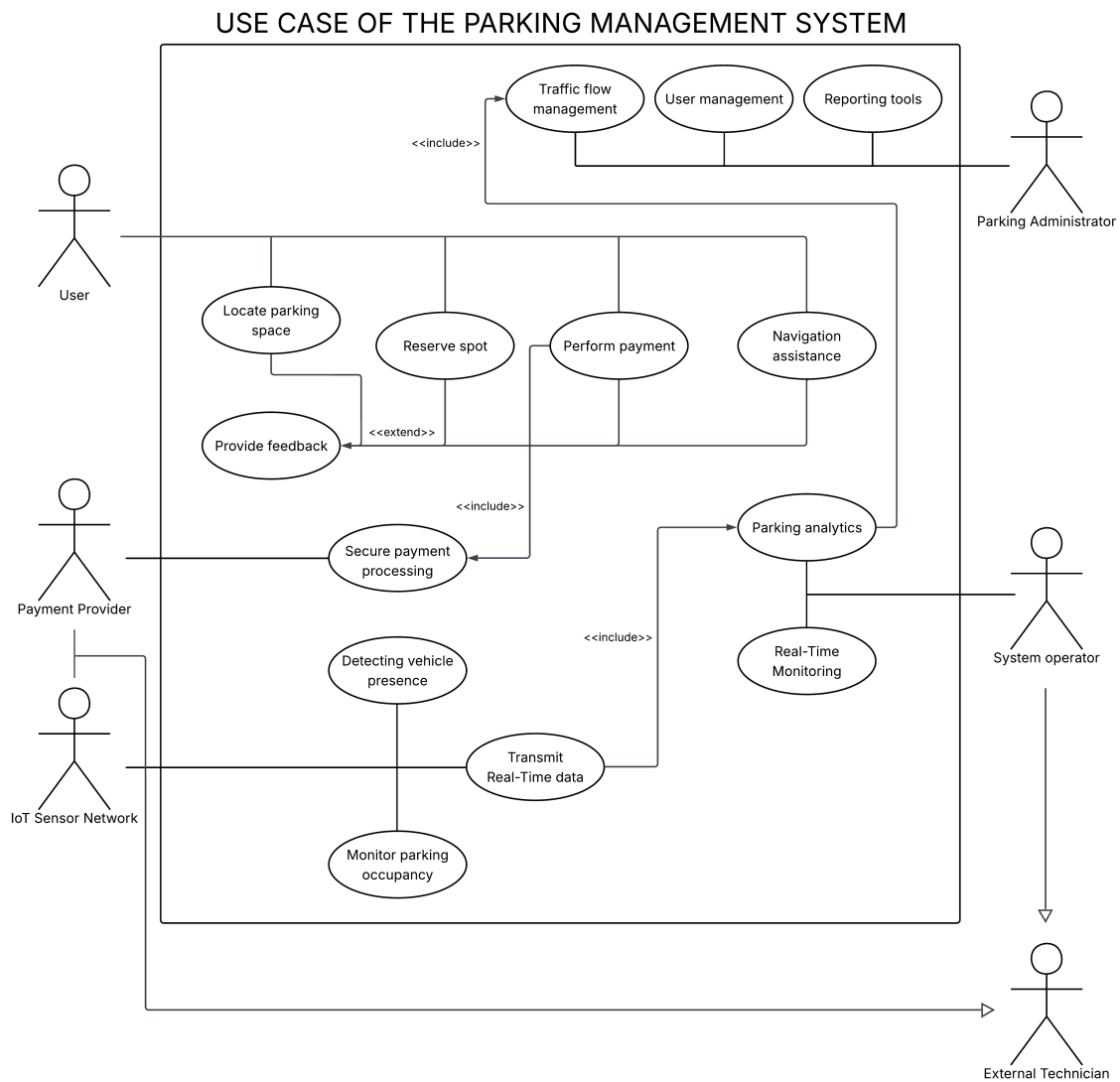
- Reservation requests.

- Payment requests.
- Parking status queries.

All communications shall be encrypted and authenticated to ensure data integrity and security. Data sharing between modules shall be designed in a modular way to ensure maintainability and scalability.

6 Use-Case Diagram

The use-case diagram presented below illustrates the primary interactions between the Smart Parking System and its external actors. The system supports multiple stakeholders, including drivers, parking administrators, system operators, payment providers, and IoT sensor networks, each responsible for different operational aspects of the platform. The diagram provides an overview of the main functionalities of the system, such as parking reservation, payment processing, occupancy monitoring, and infrastructure management.



7 UML Class Diagram

The following UML Class Diagram illustrates the main classes of the Integrated Large-Scale Parking Management System and the relationships between them. It provides a high-level view of the system architecture, including user management, parking operations, payment services, and analytical components.

Parking System UML Class Diagram

